

Dynamic Bonding Curve

0.2.1

Smart Contract Security Assessment

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Prepared for:

Meteora

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1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers, operating systems, IoT devices, and hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple, Google, and Microsoft*, have protected digital assets valued at over **\$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **\$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.



<https://offside.io/>



<https://github.com/offsidelabs>



https://twitter.com/offside_labs



2 Executive Summary

Introduction

Offside Labs completed a security audit of *Dynamic Bonding Curve* smart contracts, starting on September 2nd, 2026, and concluding on September 7th, 2026.

Project Overview

Meteora Dynamic Bonding Curve Release 0.2.1 introduces operator managed token badges for selected Token 2022 quote mints and rejects nonzero quote transfer fees during configuration, pool creation, and token transfers. It prevents new rate limiter configurations and new DAMM v1 migration pools while preserving existing pools that use those options. The release also removes deprecated protocol fee claim and zap endpoints and keeps DAMM v2 as the supported destination for new migration flows.

Audit Scope

The assessment scope contains mainly the smart contracts of the dynamic-bonding-curve program for the *Dynamic Bonding Curve* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Dynamic Bonding Curve
 - Codebase: <https://github.com/MeteoraAg/dynamic-bonding-curve>
 - Release 0.2.1: [PR-202](#)
 - Commit Hash: 2b667f5780ffe2de2f90481d1e2f3a5af67c4781

The issues described in this report have been resolved by the following commit:

- Dynamic Bonding Curve
 - Codebase: <https://github.com/MeteoraAg/dynamic-bonding-curve>
 - Commit Hash: e68871bfba80462cc2373567f146b0fd2bc857ed

We listed the files we have audited below:

- Dynamic Bonding Curve Release 0.2.1
 - programs/dynamic-bonding-curve/src/base_fee/mod.rs
 - programs/dynamic-bonding-curve/src/constants.rs
 - programs/dynamic-bonding-curve/src/error.rs
 - programs/dynamic-bonding-curve/src/event.rs
 - programs/dynamic-bonding-curve/src/instructions/admin/ix_create_operator_account.rs
 - programs/dynamic-bonding-curve/src/instructions/creator/ix_claim_creator_trading_fee.rs
 - programs/dynamic-bonding-curve/src/instructions/creator/ix_withdraw_creator_surplus.rs



- programs/dynamic-bonding-curve/src/instructions/initialize_pool/ix_initialize_virtual_pool_with_spl_token.rs
- programs/dynamic-bonding-curve/src/instructions/initialize_pool/ix_initialize_virtual_pool_with_token2022.rs
- programs/dynamic-bonding-curve/src/instructions/initialize_pool/ix_initialize_virtual_pool_with_token2022_transfer_hook.rs
- programs/dynamic-bonding-curve/src/instructions/initialize_pool/process_initialize_virtual_pool_with_token2022.rs
- programs/dynamic-bonding-curve/src/instructions/migration/dynamic_amm_v2/migrate_damm_v2_initialize_pool.rs
- programs/dynamic-bonding-curve/src/instructions/migration/ix_withdraw_migration_fee.rs
- programs/dynamic-bonding-curve/src/instructions/operator/ix_claim_protocol_fee2.rs
- programs/dynamic-bonding-curve/src/instructions/operator/ix_close_token_badge.rs
- programs/dynamic-bonding-curve/src/instructions/operator/ix_create_token_badge.rs
- programs/dynamic-bonding-curve/src/instructions/operator/mod.rs
- programs/dynamic-bonding-curve/src/instructions/partner/create_config/ix_create_config.rs
- programs/dynamic-bonding-curve/src/instructions/partner/create_config/ix_create_transfer_hook_config.rs
- programs/dynamic-bonding-curve/src/instructions/partner/create_config/process_create_config.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_claim_partner_trading_fee.rs
- programs/dynamic-bonding-curve/src/instructions/partner/ix_withdraw_partner_surplus.rs
- programs/dynamic-bonding-curve/src/instructions/swap/process_swap.rs
- programs/dynamic-bonding-curve/src/lib.rs
- programs/dynamic-bonding-curve/src/params/fee_parameters.rs
- programs/dynamic-bonding-curve/src/state/config.rs
- programs/dynamic-bonding-curve/src/state/mod.rs
- programs/dynamic-bonding-curve/src/state/operator.rs
- programs/dynamic-bonding-curve/src/state/token_badge.rs
- programs/dynamic-bonding-curve/src/utils/bits.rs
- programs/dynamic-bonding-curve/src/utils/mod.rs
- programs/dynamic-bonding-curve/src/utils/token.rs

Findings

The security audit revealed:

- 0 critical issue



- 0 high issue
- 1 medium issues
- 2 low issues
- 1 informational issues

Further details, including the nature of these issues and recommendations for their remediation, are detailed in the subsequent sections of this report.



3 Summary of Findings

ID	Title	Severity	Status
01	Missing DAMM v2 Quote Mint Badge Forwarding Can Block Migration	Medium	Acknowledged
02	Expired Transfer Fee Configuration Can Prevent Pool Operations	Low	Fixed
03	Token Badges Can Admit Unsupported Transfer Hook Quote Mints	Low	Acknowledged
04	Unused Error Is Not Marked as Deprecated	Informational	Fixed



4 Key Findings and Recommendations

4.1 Missing DAMM v2 Quote Mint Badge Forwarding Can Block Migration

Severity: Medium

Status: Acknowledged

Target: Smart Contract

Category: DoS

Description

A DBC badge only determines whether a quote mint can be used to create a DBC config and pool. Once the pool exists, DAMM v2 independently decides whether the mint is supported, whether its config skips mint validation, or whether a DAMM v2 badge is required.

The four deterministic migration cases are:

DAMM v2 directly supports quote mint	Config mint validation	skips validation	DAMM v2 quote badge required	Expected	Current
Yes	No		No	Success	Success
No	Yes		No	Success	Success
No	No		No	Failure	Failure
No	No		Yes	Success	Failure

The last row is the issue. The current DBC migration reads the DAMM config from `remaining_accounts[0]`, but does not forward the supplied DAMM quote mint badge to either DAMM pool initialization CPI. Therefore, DAMM v2 reports a missing or invalid badge even though the required badge exists and was provided.

```
124 /// System program.  
125 pub system_program: Program<'info, System>,  
126 // Remaining accounts:  
127 // 0. [READ-ONLY] damm v2 config account  
128 }
```

[programs/dynamic-bonding-curve/src/instructions/migration/dynamic_amm-v2/migrate_damm_v2_initialize_pool.rs#L124-L128](#)

A DBC badge is relevant only to the earlier DBC admission step. For example, a zero-fee `TransferFeeConfig` mint requires a DBC badge but is directly supported by DAMM v2, while a `PermanentDelegate` mint requires both programs' approval.

Impact

A pool can complete its DBC bonding curve but fail to migrate even when the required DAMM v2 badge exists. The transaction rolls back and can be retried, but migration re-



mains blocked until a compatible permissioned config is provided or DBC supports forwarding the badge.

Recommendation

It is recommended to accept an optional DAMM v2 quote-mint badge and forward it to both DAMM pool initialization CPI paths while preserving the existing no-badge path. DAMM v2 should perform the final mint, permission, and badge validation.

If mint validation fails, an DAMM v2 operator can create either a compatible DAMM v2 config with `CreatePoolWithoutMintValidation` or the required token badge, after which migration can be retried.

Mitigation Review Log

Meteora Team: All DAMM v2 configs used for DBC migrations will have `CreatePoolWithoutMintValidation`. Unsupported quote mints will instead be reviewed and whitelisted by the DBC operator.

Offside Labs: So once all DAMM v2 migrations use configs with the `CreatePoolWithoutMintValidation` permission, validation of unsupported quote mint is delegated from DAMM v2 to the dbc operator when creating the dbc token badge.

4.2 Expired Transfer Fee Configuration Can Prevent Pool Operations

Severity: Low

Status: Fixed

Target: Smart Contract

Category: Logic Error

Description

The `TransferFee` validation requires both `older_transfer_fee` and `newer_transfer_fee` to have zero basis points without considering their activation epochs.

```
174     let older_transfer_fee_bps = u16::from(  
175         transfer_fee_config  
176             .older_transfer_fee  
177             .transfer_fee_basis_points,  
178     );  
179     let newer_transfer_fee_bps = u16::from(  
180         transfer_fee_config  
181             .newer_transfer_fee  
182             .transfer_fee_basis_points,  
183     );  
184     return Ok(older_transfer_fee_bps == 0 && newer_transfer_fee_bps ==  
        0);
```



[programs/dynamic-bonding-curve/src/utils/token.rs#L174-L184](#)

Token-2022 only uses `older_transfer_fee` before `newer_transfer_fee.epoch` . After that epoch, `older_transfer_fee` is historical and no longer affects transfers.

```
152     pub fn get_epoch_fee(&self, epoch: u64) -> &TransferFee {
153         if epoch >= self.newer_transfer_fee.epoch.into() {
154             &self.newer_transfer_fee
155         } else {
156             &self.older_transfer_fee
157         }
158     }
```

[interface/src/extension/transfer_fee/mod.rs#L152-L158](#)

Consequently, when `newer_transfer_fee` has become an active zero fee configuration while the historical `older_transfer_fee` remains `nonzero` , DBC rejects the mint even though its effective transfer fee is zero. If the TransferFee authority has already been revoked, the mint cannot issue another zero-fee update to clear the historical value.

Impact

An existing pool may be unable to swap, transfer quote fees or surplus, or migrate despite its effective transfer fee being zero. The restriction can become permanent if the transfer-fee authority has been revoked.

Recommendation

It is recommended to evaluate the fee configurations according to the current epoch.

- Before `newer_transfer_fee.epoch` , require both the active older fee and scheduled newer fee to have zero basis points;
- Once the newer fee is active, require only the newer fee to be zero.

Alternatively, validating only the transfer fee effective in the current epoch by using `TransferFeeConfig::get_epoch_fee` , rather than requiring both stored fee schedules to be zero.

```
let current_epoch = Clock::get()?.epoch;
let newer_epoch =
    u64::from(transfer_fee_config.newer_transfer_fee.epoch);

if current_epoch < newer_epoch {
    return Ok(older_fee_bps == 0 && newer_fee_bps == 0);
}

Ok(newer_fee_bps == 0)
```



Mitigation Review Log

Fixed in the commit 065061c9bfb2495badb38b0f4f8f7d7d29e32fcb.

4.3 Token Badges Can Admit Unsupported Transfer Hook Quote Mints

Severity: Low

Status: Acknowledged

Target: Smart Contract

Category: Logic Error

Description

Token-2022 quote mints containing `TransferHook` or `NonTransferable` are considered unsupported by `is_supported_quote_mint` and can therefore receive a token badge. The badge then allows config and pool creation to succeed.

```
226 let extensions = mint.get_extension_types()?;  
227 for e in extensions {  
228     if e != ExtensionType::MetadataPointer && e !=  
229         ExtensionType::TokenMetadata {  
230         return Ok(false);  
231     }  
}
```

[programs/dynamic-bonding-curve/src/utils/token.rs#L226-L231](#)

However, DBC does not accept Transfer Hook accounts for quote token transfers, and Non-Transferable tokens cannot be transferred at all. Since every swap transfers the quote token, pools created with these mints cannot execute swaps.

Impact

A config and pool can be created successfully but cannot trade or reach the migration threshold through the normal lifecycle.

Recommendation

It is recommended to reject token badge creation for quote mints containing `TransferHook` or `NonTransferable`. Token badges should only bypass the default mint allowlist for extensions that remain compatible with all quote token transfer paths.

Mitigation Review Log

Meteora Team: We will only create badges for xStock or mints we know.



4.4 Informational and Undetermined Issues

Unused Error Is Not Marked as Deprecated

Severity: Informational

Status: Fixed

Target: Smart Contract

Category: Code QA

`InvalidQuoteMint` is no longer referenced by the program but remains defined without a deprecation marker. It's recommended to mark it as deprecated to preserve its error-code position while making its obsolete status explicit.

Mitigation Review Log

Fixed in the commit [83da5e0896fa67485de478d29e6db4d89a53526f](#).



5 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

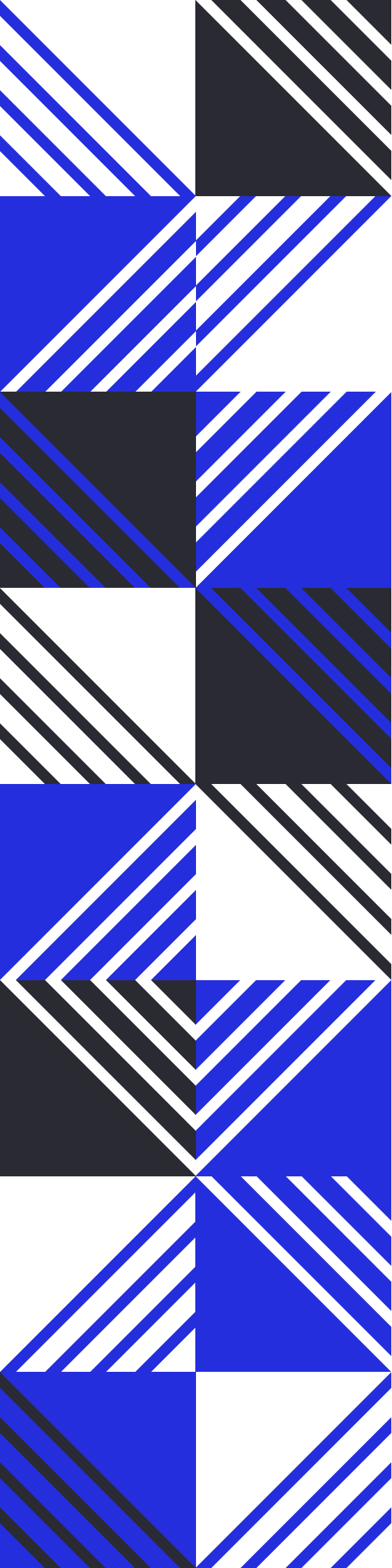
We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.



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